Reinforcement Learning-Based Optimal Setpoint Determination in Energy Utility Systems

Master Thesis

Master of Science in Artificial Intelligence

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Task description

Energy utility systems are the backbone of modern energy infrastructure. Their complexity ranges from small, localized systems to vast, highly interconnected infrastructure networks. Most of them operate under dynamic conditions where multiple interacting variables influence operational efficiency, cost, reliability, and stability. Conventionally, optimal setpoints for such systems are determined using mathematical optimization. These approaches rely on accurate system models and are typically formulated as constrained optimization problems. While effective for well-defined and moderately sized systems, classical optimization becomes computationally expensive as system complexity increases. Furthermore, time-varying dynamics, uncertainty from the integration of renewable sources, and model-plant mismatches makes it challenging to use purely model-based approaches.

This thesis investigates whether reinforcement learning can find optimal operating setpoints for two case studies of different complexity: a battery energy storage system and a non-linear cookie production factory. For each system, the control task is formulated as a Markov decision process (MDP) with explicit state, action, and reward definitions aligned with operational objectives. The RL agent is trained to produce adaptive setpoint policies that maximize battery profit or minimize cookie-factory operating cost while satisfying process constraints and preserving stable operation. The learned policies are then benchmarked against established mathematical optimization methods to determine where RL provides practical benefits and where its limitations remain.

The evaluation of the proposed method will be conducted through systematic simulation studies on two energy system case studies of varying complexity. Performance will be assessed using quantitative metrics including economic performance, constraint violation frequency, and computational time. The results will provide a structured comparison between reinforcement learning and classical optimization and will determine under which conditions RL offers practical advantages for real-time setpoint optimization in complex energy utility systems.

Abstract

Example summary in English.

Zusammenfassung

Beispielzusammenfassung in Deutsch.

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List of Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
ANN	Artificial Neural Network
BESS	Battery Energy Storage System
CPU	Central Processing Unit
EUR	Euro
HTHP	High-Temperature Heat Pump
IPOPT	Interior Point OPTimizer
KKT	Karush–Kuhn–Tucker
MDP	Markov Decision Process
MLP	Multilayer Perceptron
NLP	Nonlinear Programming
RL	Reinforcement Learning
SAC	Soft Actor-Critic
SG	Steam Generator
SOC	State of Charge
TD3	Twin Delayed Deep Deterministic Policy Gradient
TES	Thermal Energy Storage
WT	Wind Turbine

1 Introduction

The operation of modern energy utility systems is shaped by dynamic conditions and multiple sources of uncertainty. Electricity prices vary over time, renewable generation is weather dependent, and storage systems couple current decisions with future operating flexibility [1]. At the same time, industrial energy systems often contain nonlinear component behavior, operating and coupling constraints that must be respected at every time step. The resulting setpoint selection problem is therefore not only an economic problem, but also a constrained sequential decision-making problem.

In many engineering applications, such setpoints are determined using mathematical optimization. This approach is attractive because physical constraints and objective functions can be formulated explicitly, and mature solvers can compute high-quality solutions for well-defined models. For energy systems, nonlinear programming methods such as IPOPT [2] are commonly used when component models, storage dynamics, and operational constraints are nonlinear [3]. In a real-plant application, Kyriakidis, Kansara, and Roldán Serrano show that deterministic and stochastic optimization methods can have high computational costs and slow convergence for large-scale nonlinear problems with complex constraints. They further show that forecast uncertainty, optimization-horizon length, and the available computation time affect operational decisions.[4].

Reinforcement learning (RL) provides a different perspective. Unlike classical control and optimization methods that solve a trajectory-optimization problem from scratch at each operating horizon, RL agents learn a policy or action-value function gradually through interactive experience. Once trained, this learned representation enables rapid selection of actions from the current observation [5]. This makes RL attractive for real-time or repeated decision-making problems, especially when the system is complex, nonlinear, and operates in a changing environment [6]. Nevertheless, RL does not automatically guarantee feasibility or optimality. For engineering use, a learned policy must be evaluated not only by its reward, but also by its operating cost, constraint satisfaction, and computation time relative to established optimization methods.

This thesis investigates whether RL can be used for optimal setpoint determination in energy utility systems. Two case studies are considered. The first is a battery energy storage system (BESS), which represents a comparatively low-dimensional storage-control problem with time-varying electricity prices. The second is a nonlinear cookie production factory with an electrified steam-generation utility system, wind-power availability, thermal energy storage, and coupled process constraints based on the benchmark systems described by Walden et al. and Kyriakidis et al. [3, 7].

The central research question of this thesis is:

> *Can reinforcement learning determine feasible and cost-effective operating setpoints for energy utility systems of varying complexity?*

This question is addressed through the following subquestions:

- How can the setpoint optimization problem be formulated as a Markov decision process with suitable states, actions, and rewards?
- Can deep reinforcement-learning policies learn continuous setpoint decisions that respect the system dynamics and operational constraints of the two systems?

- How does the learned policy compare with mathematical optimization in terms of operating cost, constraint satisfaction, and computation time?
- Under which conditions does RL offer a practical advantage, and where do classical optimization methods remain superior?

For each case study, the setpoint optimization problem is first formulated as a Markov decision process (MDP). Deep neural networks are used to parameterize the RL policies that map system observations to continuous setpoint decisions. The resulting policies are evaluated through deterministic rollouts in reinforcement-learning environments, and the resulting schedules are compared with mathematical optimization baselines. The comparison is based on total grid-electricity cost, constraint violations, and computation time. This separation is important because a policy with high cumulative reward is only useful if it remains feasible and economically meaningful under the original engineering objective.

The remainder of this thesis is structured as follows. Section 2 establishes the theoretical foundations of setpoint optimization and reinforcement learning, and introduces mathematical optimization as the benchmark used to evaluate the RL policies. It also describes the deep RL methods considered in this work. Section 3introduces the two case studies: a battery energy storage system and a nonlinear cookie production factory. Section 4 presents the methodological framework, covering the MDP formulation, environment design, policy-training procedure, optimization baselines, and evaluation metrics. Section 5 reports and discusses the experimental results, with particular emphasis on the comparison between RL policies and mathematical optimization. Finally, Section 6 summarizes the main findings, identifies the limitations of the study, and outlines directions for future work.

2 Theory and Background

This section provides the theoretical basis for the setpoint optimization problem studied in this thesis. The central question is how operating setpoints in energy utility systems can be selected over time so that costs are reduced while physical and operational limits are respected. The same problem can be viewed from two perspectives: as a model-based constrained optimization problem, and as a sequential decision-making problem that can be addressed with reinforcement learning. The section first introduces the fundamental concepts of RL and the MDP formulation [5], then presents deep RL methods for continuous control and the TD3 [8] and SAC [9] algorithms evaluated in this work.

Here, RL is not treated as a substitute for physical constraints. Instead, the setpoint-optimization problem is formulated as an MDP, with feasibility enforced where possible through action bounds, state-dependent mappings, and system dynamics, and otherwise discouraged through reward penalties. This distinction is important when comparing RL with mathematical optimization, because a low-cost policy is only useful if its decisions remain feasible.

2.1 Setpoint Optimization in Energy Utility Systems

Energy utility systems are operated by choosing setpoints for controllable components such as storage units, heat pumps, valves, and bypasses. A setpoint describes the desired value of a control variable. In the systems considered here, this may be the charging or discharging power of a battery, the bypass variable β_1, or a mass flow rate. These choices are not just technical details; they directly affect operating cost, energy efficiency, constraint satisfaction, and overall system reliability and lifetime [10].

Setpoint determination is also inherently sequential. The best operating point at one time step cannot be chosen independently of what happened before or what may be needed later. Storage systems make this especially clear. Charging a battery or a thermal energy storage can increase future flexibility, but it may also increase the current electricity cost. Discharging can reduce current grid consumption or generate sale revenue, but it leaves less stored energy for later. Good setpoint decisions must therefore account for inter-temporal dependencies and physical operating limits [3]. Each action affects both the immediate economic outcome and the feasible decisions available at later time steps; consequently, its quality must be evaluated over the full planning horizon rather than solely at the current time step [11].

In practical operation, the controller also has to react to external signals such as electricity prices, renewable generation, production demand, and ambient conditions. Since these signals change over time, fixed rule-based strategies can become inefficient, and the operation can be formulated as a finite-horizon decision problem with time-varying boundary conditions [12, 6]. This motivates two formulations of the same setpoint determination task: constrained finite-horizon optimization and reinforcement learning.

2.2 Mathematical Optimization

Mathematical optimization provides a natural way to describe setpoint selection. In general, optimization means selecting the best decision from a set of feasible alternatives according to a defined objective. For energy systems, the objective is often to minimize operating cost or energy consumption, while the feasible set

is determined by physical equations and operational limits [3, 4]. A constrained optimization problem can be written as [13]:

$$\min_{x\in\Omega} f(x) \quad \text{s.t.} \quad h(x) = 0,\ g(x) \le 0, \tag{1}$$

Here, $x \in \Omega \subseteq \mathbb{R}^n$ denotes the decision variables, $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function, $h : \mathbb{R}^n \to \mathbb{R}^p$ represents equality constraints, and $g : \mathbb{R}^n \to \mathbb{R}^q$ represents inequality constraints. The solution $x^\star$ is optimal if it minimizes the objective while satisfying all constraints.

For finite-horizon setpoint optimization, the decision vector contains the control actions for several time steps. The objective is usually the accumulated operating cost over the horizon. The constraints describe system dynamics, energy balances, storage bounds, and terminal requirements. This time-coupled structure is important because a state such as battery state of charge or thermal storage temperature depends on previous decisions.

The difficulty of an optimization problem depends strongly on its mathematical structure. Linear and mixed-integer linear formulations are widely used in energy-system applications because simplified models can often be solved reliably. However, realistic energy conversion systems frequently include nonlinear component behavior, continuous control variables, and coupled dynamic constraints. Such systems lead to nonlinear programming (NLP) formulations, which are generally harder to solve and may contain local optima [13]. As the number of time steps, decision variables, nonlinear component models, and coupling constraints increases, the optimization problem generally becomes larger and more difficult to solve within a short operational time window [7].

Another important distinction is whether the problem is deterministic or stochastic. In deterministic optimization, all relevant parameters, such as electricity prices, wind availability, and demand, are assumed to be known over the optimization horizon. Stochastic optimization represents uncertain quantities through random variables or scenarios. This is closer to real operation, but it also increases model size and computational complexity.

Nonlinear constrained problems are usually solved with dedicated numerical solvers. Interior-point methods are common in this setting. Wächter and Biegler introduce IPOPT as an open-source solver for large-scale nonlinear programming that uses a primal-dual interior-point method with a filter line-search strategy [2].

To illustrate the interior-point principle, consider the general problem in Eq. (1). Introducing slack variables converts the inequality constraints into $g(x) + s = 0$, where $s \ge 0$. A logarithmic barrier with parameter $\mu > 0$ restricts the iterates to the interior, $s > 0$, and leads to the subproblem

$$\begin{aligned} \min_{x,s} \quad & f(x) - \mu \sum_{j=1}^{q} \ln(s_j) \\ \text{s.t.} \quad & h(x) = 0, \qquad g(x) + s = 0, \qquad s > 0. \end{aligned} \tag{2}$$

The logarithmic barrier prevents the iterates from reaching the constraint boundary. IPOPT solves a sequence of such subproblems using a primal-dual Newton method. With equality multipliers y and inequality multipliers z, the perturbed Karush–Kuhn–Tucker conditions are

$$\begin{aligned} \nabla f(x) + J_h(x)^\top y + J_g(x)^\top z = 0, \\ h(x) = 0, \qquad g(x) + s = 0, \\ Sz = \mu \mathbf{1}, \qquad s > 0,\ z > 0, \end{aligned} \tag{3}$$

where J_h and J_g are the Jacobians of the equality and inequality constraints, respectively, and $S = \mathrm{diag}(s)$. As μ is reduced towards zero, the perturbed complementarity condition $Sz = \mu\mathbf{1}$ approaches the KKT complementarity condition of the original constrained problem [2].

Nocedal and Wright note that the performance of nonlinear optimization algorithms depends on formulation quality, particularly the constraint structure, variable scaling, and initial point [13].

These considerations motivate the investigation of reinforcement learning as a complementary approach to model-based optimization. The following section introduces its fundamental principles and its formulation for setpoint optimization.

2.3 Reinforcement Learning

Reinforcement learning (RL) offers a different view of the same setpoint problem. Instead of solving an optimization problem from scratch at every decision time, an agent learns from repeated interaction with an environment. At each time step, the agent observes the current state, selects an action, receives a reward, and observes the next state. Over time, it adapts its policy so that actions leading to high long-term reward become more likely [5].

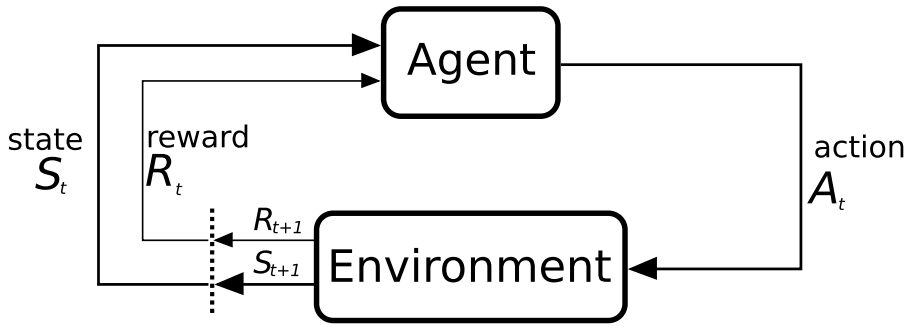


Figure 1: Interaction loop between the reinforcement-learning agent and the environment in a Markov decision process [14].

Formally, RL problems are commonly modeled as a Markov Decision Process (MDP), defined by the tuple [5]

$$(\mathcal{X}, \mathcal{A}, p, r, \gamma_{\mathrm{RL}}),$$

where

- $\mathcal{X}$ is the set of states representing the system,
- $\mathcal{A}$ is the set of actions,
- $p(x' \mid x, a)$ defines the transition dynamics,
- $r(x, a)$ is the reward function,
- $\gamma_{\mathrm{RL}} \in [0, 1]$ is the discount factor.

Under the Markov property, the current state contains all information required to predict the next state, given the selected action. A policy $\pi(a \mid x)$ describes how the agent selects actions. To evaluate a decision, RL considers both its immediate reward and the rewards obtained thereafter. For a finite horizon of N decision steps, this quantity, called the return, is

$$G_t = \sum_{i=0}^{N-t-1} \gamma_{\mathrm{RL}}^i r(x_{t+i}, a_{t+i}),$$

and the learning objective is to find a policy that maximizes the expected return. The value function and action-value function of a policy π are defined as

$$v_t^\pi(x) = \mathbb{E}_\pi[G_t \mid X_t = x], \qquad q_t^\pi(x,a) = \mathbb{E}_\pi[G_t \mid X_t = x, A_t = a].$$

The value function estimates the expected return from a state, whereas the action-value function estimates the expected return from a state-action pair. These estimates allow the agent to consider the future consequences of a decision, not only its immediate reward.

For a finite-horizon problem, the optimal action-value function can be written recursively as

$$q_t^\star(x,a) = r(x,a) + \gamma_{\mathrm{RL}}\, \mathbb{E}\big[v_{t+1}^\star(x') \mid x, a\big].$$

Here, x' denotes the next state. Rather than calculating this expectation from a known system model, RL learns an approximation of the action-value function from sampled transitions (x_t, a_t, r_t, x_{t+1}). The policy is then improved to select actions with higher estimated long-term return. This illustrates the central idea of RL: actions are evaluated not only by their immediate reward, but also by their expected future consequences [5].

Classical dynamic programming can solve MDPs through backward recursion. However, this approach becomes impractical when the state or action space grows, a difficulty known as the curse of dimensionality [5]. Realistic energy systems often have continuous states, continuous actions, and long horizons, so exact dynamic-programming solutions are often computationally infeasible.

RL addresses this limitation by learning approximate policies from interaction data. A trained policy can then compute an action directly from the current observation. In model-free RL, the learning algorithm does not require an explicit analytical transition model for planning, although training still requires interaction data. In engineering applications, this data is usually generated with a simulator or digital model, so the quality of the learned policy depends strongly on the quality of the training environment.

To locate the algorithm used in this thesis, it is helpful to briefly classify the main families of RL algorithms. RL methods are not separated by one single list of types; instead, they are usually described through several overlapping distinctions:

- Model-based and model-free methods: model-based methods learn or use a model of the environment dynamics for planning. Model-free methods learn directly from sampled transitions.

- Value-based and policy-based methods: value-based methods learn value or action-value functions and choose actions from them. Policy-based methods directly optimize a parameterized policy.

- Actor-critic methods: actor-critic methods combine ideas above. The actor represents the policy, and the critic estimates values or action-values used to improve that policy.

- On-policy and off-policy methods: on-policy methods learn from data generated by the current policy. Off-policy methods can also reuse data generated by previous policies or exploratory behavior.

- Tabular and deep RL methods: tabular methods store values explicitly for small state and action spaces. Deep RL uses neural networks as function approximators and is therefore more suitable for large or continuous spaces.

The relevant setpoints in this thesis, including battery power, temperature levels, mass flow rates, and bypass positions, are continuous quantities. Discretizing them would either exclude useful operating points or create a very large action space. TD3 and SAC are therefore selected because they learn parameterized policies that output continuous actions directly.

Both methods use an actor-critic architecture. The actor maps the current state to continuous setpoint decisions, while the critic estimates their expected long-term effect on operating cost and future system states. TD3 and SAC are applied to learn continuous setpoint policies for both case studies.

2.3.1 TD3 and SAC

Twin Delayed Deep Deterministic Policy Gradient (TD3). TD3 uses a deterministic actor, $\mu_\theta(x)$, that maps the current state directly to a continuous action. Two critics estimate the action value, and the smaller estimate is used when constructing the learning target to reduce overestimation bias. TD3 further stabilizes learning through delayed actor updates and target-policy smoothing, in which small noise is added to target actions during critic updates [8].

$$J_{\mathrm{TD3}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}}\left[q_{\phi_1}\big(x, \mu_\theta(x)\big)\right].$$

Here, $J_{\mathrm{TD3}}(\theta)$ is the actor objective; θ denotes the actor parameters; $\mathcal{D}$ is the replay buffer from which states x are sampled; $\mu_\theta(x)$ is the deterministic action proposed by the actor; and q_{ϕ_1} is the first critic, parameterized by ϕ_1, which estimates the action value. The TD3 actor is trained to maximize this objective, producing one deterministic action, $a = \mu_\theta(x)$, for each state.

Soft Actor-Critic (SAC). SAC instead uses a stochastic actor, $\pi_\theta(a \mid x)$, that defines a distribution over continuous actions. In addition to maximizing expected return, SAC includes an entropy term that encourages exploration while training. Like TD3, SAC uses two critics and experience replay; however, its stochastic policy and entropy objective distinguish it from the deterministic TD3 policy [9].

$$J_{\mathrm{SAC}}(\pi) = \mathbb{E}_\pi\left[\sum_{t=0}^{N-1} \gamma_{\mathrm{RL}}^t \left(r(x_t, a_t) + \alpha\mathcal{H}(\pi(\cdot \mid x_t))\right)\right].$$

where $J_{\mathrm{SAC}}(\pi)$ is the objective of the stochastic policy π; $\mathbb{E}_\pi$ denotes the expectation over trajectories generated by that policy; N is the number of decision steps; γ_{RL} is the discount factor; $r(x_t, a_t)$ is the reward at time t; α weights the entropy term; and $\mathcal{H}(\pi(\cdot \mid x_t))$ is the policy entropy.

For a state x, the policy entropy is defined as

$$\mathcal{H}\big(\pi(\cdot \mid x)\big) = -\mathbb{E}_{a\sim\pi(\cdot|x)}\left[\log \pi(a \mid x)\right].$$

where a is an action sampled from the policy, and $\log \pi(a \mid x)$ is the logarithm of its probability density. During training, entropy is estimated from sampled actions as $\mathcal{H}(\pi(\cdot \mid x)) \approx -\log \pi(a \mid x)$ [9].

The two algorithms therefore provide a comparison between deterministic and stochastic actor–critic policies for continuous setpoint optimization. Their performance is evaluated using operating cost, constraint satisfaction, and computation time.

3 Use Cases

3.1 Battery Energy Storage System

Battery energy storage systems (BESS) are electrochemical devices that store electrical energy for later use and provide grid services such as energy arbitrage, peak shaving, and short-term balancing. A typical BESS is characterized by its usable energy capacity, maximum charge/discharge power, round-trip efficiency, and state-of-charge (SOC) operating limits. Real-world operation must balance economic objectives (e.g., buying electricity at low prices and selling it at high prices) with engineering constraints (SOC bounds, power limits, efficiency losses).

BESS presents several challenges. Decisions are inherently time-coupled because current charging/discharging changes future availability. In addition, uncertain exogenous signals (market prices) require policies that are robust to forecast errors, and hard feasibility constraints must be enforced at every time step. These properties make BESS an illustrative case study for comparing optimization-based and reinforcement-learning approaches to real-time setpoint determination in energy systems.

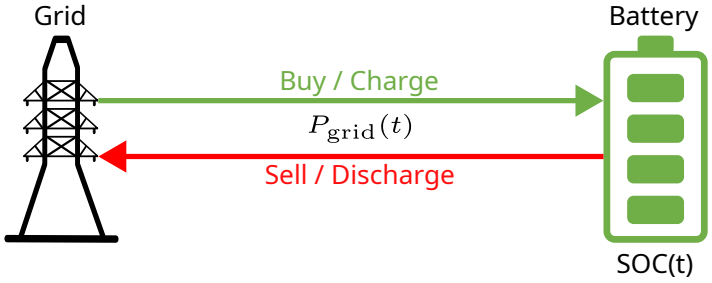


Figure 2: Schematic of the battery energy storage system. The battery charges from the grid and sells electricity back to the grid; its state of charge SOC(t) records the stored energy.

A battery energy storage system provides flexible, time-shiftable power by buying electricity when prices are low and selling it back to the grid when prices are high. This price-arbitrage operation maximizes profit. For simplicity, the same time-varying electricity price S_t is used for purchases and sales. In this thesis, the battery is modeled using the `BatteryEnv` environment: the agent's observation includes the state-of-charge (SOC), a normalized current price, a normalized time index, and the normalized price forecast for the complete scheduling horizon; the action

is a continuous charge/discharge command $u \in [-1, 1]$ scaled to grid-side power $P_t^{\mathrm{S}} \in [-P_{\max}, P_{\max}]$.

The SOC dynamics are modeled as

$$\mathrm{SOC}_{t+1} = \mathrm{SOC}_t + \frac{P_{\mathrm{actual},t}\,\Delta t}{E_{\max}},$$

where $P_{\mathrm{actual},t}$ is the effective storage-side power after efficiency losses, $E_{\max}$ the usable energy capacity, and Δt the time step. The sign convention is $P_t^{\mathrm{S}} > 0$ for charging and grid purchase, and $P_t^{\mathrm{S}} < 0$ for discharging and electricity sale. The superscript S identifies power exchanged with the electricity supply grid. The mathematical formulation uses the following smooth power and terminal constraints:

$$\begin{aligned} P_t^{\mathrm{S}} = P_{\max}\Big[&\frac{1}{2}\big(1+\tanh\big(k(u_t-\delta)\big)\big)\,u_t \\ &\times\frac{1}{2}\big(1+\tanh\big(k(0.995-\mathrm{SOC}_t)\big)\big) \\ &+\frac{1}{2}\big(1-\tanh\big(k(u_t+\delta)\big)\big)\,u_t \\ &\times\frac{1}{2}\big(1+\tanh\big(k(\mathrm{SOC}_t-0.005)\big)\big)\Big], \quad \forall t, \end{aligned} \tag{4}$$

$$\begin{aligned} P_{\mathrm{actual},t} = \frac{P_t^{\mathrm{S}}}{2}\Bigg[&\left(1+\tanh\left(\frac{P_t^{\mathrm{S}}}{\epsilon}\right)\right)\eta \\ &+\left(1-\tanh\left(\frac{P_t^{\mathrm{S}}}{\epsilon}\right)\right)\frac{1}{\eta}\Bigg], \quad \forall t, \end{aligned} \tag{5}$$

$$\mathrm{SOC}_{\mathrm{final}} \geq \mathrm{SOC}_{\mathrm{term}}. \tag{6}$$

Here, u_t is the normalized control command, $P_{\max}$ is the maximum charging and discharging power, k controls the steepness of the smooth action and SOC gates, and $\delta = 0.0025$ defines a small action deadband. The terminal-SOC requirement is $\mathrm{SOC}_{\mathrm{term}}^{\min} = 0.5$.

The parameter η describes the charging and discharging efficiency, while ϵ controls the smooth transition between the two operating directions. Equation (4) maps the normalized action and current SOC to grid-side battery power. It progressively restricts charging as SOC approaches 0.995 and discharging as SOC approaches 0.005. Equation (5) maps grid-side power to storage-side power through the smooth efficiency relation. Finally, the mathematical-optimization model enforces the terminal-SOC requirement in Eq. (6) as a hard constraint.

The battery objective is to maximize profit. The grid-side cost contribution is calculated as the electricity price multiplied by the battery power, and its minimization is equivalent to profit maximization. Accordingly, the mathematical-optimization objective is

$$\min_{\{u_t\}_{t=0}^{N-1}} J = \sum_{t=0}^{N-1} S_t P_t^{\mathrm{S}} \Delta t,$$

where S_t is the electricity price at time step t. The battery purchases electricity from the grid when prices are low and sells it back to the grid when prices are high. Main challenges include respecting SOC and power limits, accounting for efficiency losses, handling uncertain future prices, and managing the fact that actions now affect future energy availability. These features make the BESS a suitable test case

for reinforcement learning, where a policy must balance using energy now against saving it for likely higher-price periods while always obeying feasibility limits.

3.2 Cookie Production Factory

The second use case represents a non-linear cookie production factory with an electrified steam-generation utility system. The production process requires a constant supply of process heat in the form of superheated steam, while the utility system must decide how to operate the energy conversion components under fluctuating wind generation W_t and electricity prices S_t. The operational objective is to satisfy the steam demand while minimizing grid electricity cost over a finite scheduling horizon. The system model is based on the industrial power-to-heat system introduced by Walden et al. [3] and later used as a renewable steam-generation benchmark by Kyriakidis et al. [7].

The system consists of four main units: a wind turbine (WT), a closed reverse Brayton-cycle high-temperature heat pump (HTHP), a sensible thermal energy storage (TES), and a steam generator (SG). The WT supplies renewable electricity, while additional electricity can be purchased from the grid. The HTHP uses a waste-heat air stream as heat source and delivers high-temperature process heat to an intermediate thermal-oil loop. The thermal oil acts as heat transfer fluid and can be routed through the TES or directly to the SG by controllable bypasses. The SG provides the required steam conditions for the consumer factory. The HTHP and SG are represented by polynomial surrogate models, while the TES is modeled as a simplified single energy reservoir, and the charging/discharging behavior is represented using efficiency equations.

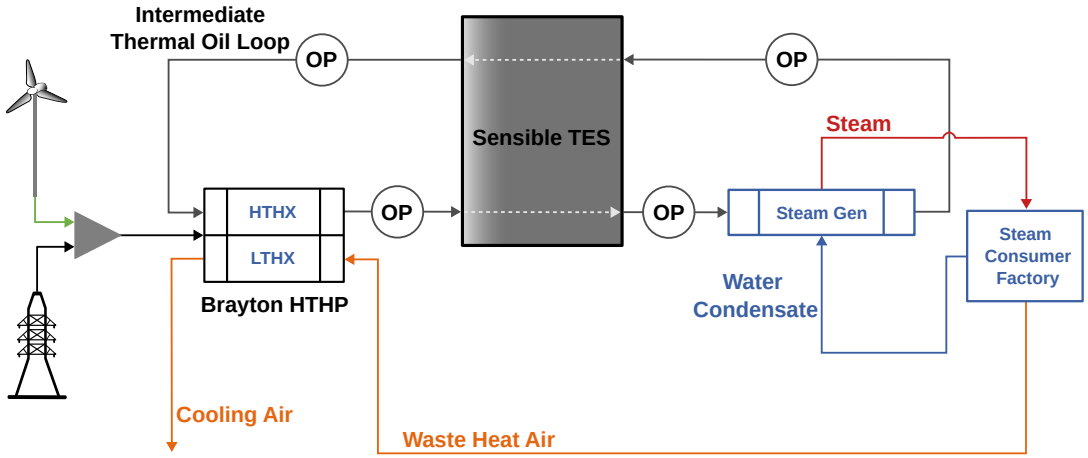


Figure 3: Schematic representation of the electrified steam-generation system serving the cookie production factory. Reproduced from Kyriakidis et al. [7]; the system was originally introduced by Walden et al. [3].

The corresponding finite-horizon optimization problem is formulated over a dis-

crete time horizon with time index $t \in \{1, \ldots, n\}$:

$$\min_{\mathbf{x}} \quad J = \sum_{t=1}^{n} P_t^{\mathrm{S}} S_t \Delta t, \tag{7}$$

$$\text{s.t.} \quad P_t^{\mathrm{S}} + W_t = 3F_{\mathrm{HTHP}}\left(T_{\mathrm{I}}^t, \dot{m}_{\mathrm{I}}^t, T_{\mathrm{III}}^t, R^t\right), \tag{8}$$

$$T_{\mathrm{II}}^t = F_{\mathrm{HTHX}}\left(T_{\mathrm{I}}^t, \dot{m}_{\mathrm{I}}^t, T_{\mathrm{III}}^t, R^t\right), \tag{9}$$

$$T_{\mathrm{IV}}^t = F_{\mathrm{LTHX}}\left(T_{\mathrm{I}}^t, \dot{m}_{\mathrm{I}}^t, T_{\mathrm{III}}^t, R^t\right), \tag{10}$$

$$T_2^t = \beta_1^t T_{\mathrm{II}}^t + \left(1 - \beta_1^t\right) T_1^t, \tag{11}$$

$$T_{\mathrm{I}}^t = \beta_2^t T_3^t + \left(1 - \beta_2^t\right) T_4^t, \tag{12}$$

$$T_2^t = 201.92 + \frac{1819.32}{3\dot{m}_{\mathrm{II}}^t}, \tag{13}$$

$$T_3^t = 196.3 - \frac{188.4}{3\dot{m}_{\mathrm{I}}^t}, \tag{14}$$

$$\dot{Q}_{s,\mathrm{ch}}^t = 3\dot{m}_{\mathrm{II}}^t c_{p,\mathrm{f}}\left(T_{\mathrm{II}}^t - T_1^t\right)\left(1 - \beta_1^t\right), \tag{15}$$

$$\dot{Q}_{s,\mathrm{dch}}^t = 3\dot{m}_{\mathrm{I}}^t c_{p,\mathrm{f}}\left(T_4^t - T_3^t\right)\left(1 - \beta_2^t\right), \tag{16}$$

$$\dot{Q}_{s,\mathrm{ch}}^t \dot{Q}_{s,\mathrm{dch}}^t \leq \gamma, \tag{17}$$

$$T_1^t = T_{\mathrm{II}}^t - \epsilon_{\mathrm{ch}}\left(T_{\mathrm{II}}^t - T_s^{t-1}\right), \tag{18}$$

$$T_4^t = T_3^t - \epsilon_{\mathrm{dch}}\left(T_3^t - T_s^{t-1}\right), \tag{19}$$

$$T_s^t = T_s^{t-1} + \frac{\dot{Q}_{s,\mathrm{ch}}^t - \dot{Q}_{s,\mathrm{dch}}^t}{m_s c_{p,\mathrm{s}}} \Delta t, \tag{20}$$

$$T_s^0 = \tilde{T}_0, \qquad T_s^n = T_s^0, \tag{21}$$

$$T_{\mathrm{I}}^t \in [177, 250], \qquad \dot{m}_{\mathrm{I}}^t \in [5, 16], \tag{22}$$

$$T_{\mathrm{III}}^t \in [60, 100], \qquad R^t \in [0.8, 1.53], \qquad \beta_1^t, \beta_2^t \in [0, 1]. \tag{23}$$

Here, S_t denotes the spot electricity price, P_t^{S} denotes grid power, and W_t denotes available wind power at time step t. The objective in Eq. (7) minimizes the electricity cost of grid power consumption. The power balance in Eq. (8) couples grid electricity, wind power, and HTHP demand. The remaining constraints describe the HTHP and SG surrogate models, bypass mixing, TES charging and discharging, the prohibition of simultaneous charge and discharge, cyclic storage operation, and the admissible operating ranges of the main decision variables.

This use case is more complex than the battery system because it combines nonlinear surrogate equations, thermal storage dynamics, multiple coupled continuous control variables, and hard process constraints. It therefore provides a suitable second benchmark for evaluating whether reinforcement learning can learn feasible and cost-effective setpoint policies for industrial energy utility systems, and for comparing the learned policy against the existing mathematical optimization formulation.

4 Methodology

The previous section introduced the two energy utility systems considered in this thesis and described their operational objectives and constraints. This section explains how these system-specific optimization problems are reformulated as reinforcement-learning problems. In each time step, the agent observes the current system state together with current and forecast external conditions, such as electricity prices

and available wind power, selects continuous control actions, and receives a reward based on economic performance and constraint satisfaction. Although the physical models and control variables differ between the two use cases, both are represented using the same agent-environment interaction structure. At each discrete time step, the environment maps the agent's action to physical setpoints, advances the system dynamics, and returns the resulting next state, reward, and observations.

4.1 Problem Formulation

For both use cases, the objective is to determine operating setpoints over a discrete time horizon that optimize the relevant economic objective while respecting all constraints. Let $t \in \{0, \ldots, N-1\}$ denote the discrete time index, S_t the spot electricity price, P_t^{S} the grid power, and W_t the available wind power. At each time step, the system is described by a state vector x_t, the agent selects a continuous action vector a_t, and the system evolves to the next state x_{t+1} according to the underlying system dynamics. For the battery case, the only external input is the electricity price S_t. For the cookie-factory case, the external inputs are the electricity price S_t and the available wind power W_t.

As introduced in Section 2, both use cases are formulated as finite-horizon Markov decision processes. The agent learns a policy that maps the current state x_t to a continuous action a_t to maximize cumulative reward. The following subsections specify the state variables, action spaces, transition dynamics, and reward functions for each system.

The engineering objectives differ between the two use cases: the battery model maximizes profit through price arbitrage, whereas the cookie factory minimizes the cost of electricity required for steam generation. The rewards are defined accordingly and are augmented by penalty terms for constraint violations where required. In both cases, feasible and economically favorable operation corresponds to a high cumulative reward. The following subsections describe how this general formulation is implemented in the reinforcement-learning framework, training setup, and evaluation procedure.

4.2 Reinforcement Learning Framework

The reinforcement-learning framework used in this thesis follows the same general structure for both use cases. The environments are implemented using Gymnasium [15], while the continuous-control policies are trained using Stable-Baselines3 [16]. Each interaction is stored as a transition

$$\tau_t = (x_t, a_t, r_t, x_{t+1}, d_t)\,,$$

where d_t indicates whether the episode has ended. In the present environments, reaching the end of the fixed scheduling horizon is represented by the `terminated` signal. The transitions are stored in a replay buffer and sampled repeatedly during training, enabling the off-policy learning used by SAC. SAC augments the conventional return-maximization objective with an entropy term during training to encourage exploration of the continuous action space. This implementation follows the general agent–environment interaction illustrated in Figure 1, extended by forecast-augmented states, environment-specific action mappings, and constraint handling.

During training, an action sampled from the stochastic actor is first processed by an environment-specific action map

$$a_t^{\text{phys}} = \mathcal{M}(x_t, a_t),$$

where $\mathcal{M}$ maps the normalized policy output to physical setpoints and applies environment-specific feasibility handling. Hard constraints are enforced through action bounds, state-dependent mappings, or the system dynamics, while requirements that are not enforced directly are represented by penalty terms in the reward.

For the battery system, the state supplied to the agent consists of the current state of charge, the normalized current spot electricity price S_t, the normalized position within the scheduling horizon, and the normalized 24-step price forecast $S_{0:N-1}$. As future work, a rolling-horizon implementation could update the forecast at every decision step. Here, the complete forecast is kept fixed to enable a direct comparison with mathematical optimization over the same single scheduling horizon. The continuous scalar action $u_t \in [-1, 1]$ represents the battery charging or discharging command: positive values correspond to charging and negative values to discharging. A smooth, state-dependent action map converts this normalized command into grid-side battery power while progressively restricting charging and discharging near the upper and lower SOC limits.

The reward has the same general form in both environments:

$$r_t = -C_t - \sum_{\ell} \lambda_\ell v_{\ell,t} - \lambda_{\text{T}} v_{\text{T},t},$$

where C_t is the grid-electricity cost at time step t, $v_{\ell,t}$ are constraint-violation terms, and λ_ℓ are their corresponding penalty weights. The terminal term $v_{\text{T},t}$ is only active at the end of an episode, and λ_{T} is its penalty weight. In the battery case it penalizes an insufficient terminal SOC, while in the cookie factory it penalizes deviation from the cyclic terminal TES temperature. This reward design converts the case-specific economic objective into a return-maximization problem without removing the need for physical feasibility.

More specifically, the battery reward implemented in the environment is

$$r_t = -\frac{S_t P_t^{\text{S}} \Delta t}{1000} - \mathbf{1}_{\{t=N-1\}} \lambda_{\text{SOC}} \max\left(SOC_{\min}^{\text{terminal}} - SOC_N, 0\right),$$

where S_t is the spot electricity price in EUR/MWh, P_t^{S} is the grid-side battery power in kW, and Δt is the step duration in hours. The factor 1000 converts the energy charge to euros. The terminal term is applied only at the end of an episode and penalizes a final SOC below the prescribed minimum. Intermediate SOC feasibility is handled directly by the battery dynamics and state-dependent power mapping rather than by a constraint-violation penalty.

As an additional feasibility measure, a progressive terminal-SOC reachability penalty was investigated. At each time step, the environment calculates the minimum SOC required to retain a feasible path to the terminal-SOC target under the maximum possible future charging rate:

$$\text{SOC}_t^{\text{reach}} = \max\left(\text{SOC}_{\min}^{\text{terminal}} - (N-t)\frac{\eta P_{\max} \Delta t}{E_{\max}}, 0\right).$$

Whenever the current SOC falls below this reachability threshold, the shaped reward is

$$r_t^{\text{reach}} = r_t - \lambda_{\text{reach}} \max\left(\text{SOC}_t^{\text{reach}} - \text{SOC}_t, 0\right),$$

where $\lambda_{\text{reach}} > 0$ weights the reachability penalty. Unlike the terminal penalty described above, this penalty provides feedback before the final time step and encourages the policy to preserve sufficient energy for the terminal requirement. The resulting feasibility–cost trade-off is reported in Section 5.

After training, both policies are evaluated deterministically to compare schedules. For SAC, the mean action of the stochastic policy is selected at each time step. TD3 directly outputs a deterministic action. This produces one reproducible setpoint trajectory for a given price and wind profile and allows direct comparison with the mathematical-optimization baseline in terms of cost, constraint satisfaction, and computation time.

4.3 Training Setup

Training is performed in reinforcement-learning environments before the learned policies are evaluated on fixed test trajectories. The environment provides the transition dynamics, cost calculation, constraint handling, and terminal condition for each episode. The learning algorithm only receives observations, rewards, and termination signals through the environment interface.

For both use cases, one episode corresponds to a 24-step scheduling horizon. During training, the start index of the episode is sampled randomly from the available time-series data. This random-window strategy exposes the agent to different price and renewable-generation patterns instead of repeatedly starting each episode at the same time of day. Both environments use the grid-price series; the cookie factory environment additionally samples the corresponding wind-availability values for the same 24-hour window. The forecast horizon is also set to 24 steps, so every observation contains the current operating state together with the complete episode profiles of the relevant exogenous signals.

The training environments are wrapped with monitoring and vectorized environment interfaces. Monitoring records episode rewards and lengths for later inspection, while the vectorized wrapper provides the interface expected by the Stable-Baselines3 implementation of SAC and TD3. Only one environment instance is used at a time in the current setup, so the vectorization is mainly an interface layer rather than a parallel simulation strategy.

The BESS reward is the negative grid-side electricity cost at each time step, which is equivalent to profit, with an additional terminal penalty if the final SOC falls below the desired terminal level. The cookie factory reward also starts from the negative grid-electricity cost, but includes additional penalties for different constraint violations. The penalty terms are therefore more important in the second use case, because feasibility cannot be represented by simple action bounds alone.

The main SAC and TD3 hyperparameters used in the implementation are summarized in Table 1. The same learning-rate schedule and discount factor are used for both use cases to provide a consistent baseline configuration. The learning-rate schedule was selected through preliminary experiments; the comparison results are reported in Section 5. Differences between the two training setups mainly arise from the different state variables, action spaces, and constraint structures of the two environments.

Both SAC policies use three hidden layers of 256 neurons with ReLU activation functions for both actor and critic networks. TD3 uses the Stable-Baselines3 default architecture of two ReLU hidden layers with 400 and 300 neurons for both actor and critic networks. In all cases, trained models are saved after learning and later

Parameter	Battery system	Cookie factory
Algorithm	SAC and TD3	SAC and TD3
Policy	MLP policy	MLP policy
Network architecture	SAC:3×256; TD3:400,300 ReLU	SAC:3×256; TD3:400,300 ReLU
Replay buffer size	200000	200000
Batch size	256	256
Learning rate	linear schedule from $3 \cdot 10^{-4}$	linear schedule from $3 \cdot 10^{-4}$
Discount factor γ_{RL}	0.99	0.99
Target smoothing coefficient τ	0.005	0.005
Episode length	24 steps	24 steps

Table 1: Main SAC and TD3 training settings used for the two reinforcement-learning environments. TD3 uses the Stable-Baselines3 default network architecture; the remaining listed settings are shared.

loaded for deterministic evaluation. This separation between training and evaluation is important: stochastic actions are useful during SAC learning, but deterministic actions are used for the final comparison with mathematical optimization.

4.4 Evaluation Metrics

The trained RL policies are evaluated through deterministic rollouts on fixed 24-hour evaluation windows. Deterministic evaluation means that the actor returns one action for each observation without sampling exploration noise. This makes the resulting setpoint trajectory reproducible and allows a direct comparison with the mathematical optimization baseline on the same grid-price and wind-availability profiles.

The primary economic metric is the total net grid-side cost over the evaluation horizon:

$$J_{\mathrm{grid}} = \sum_{t=0}^{N-1} C_t,$$

where C_t is the grid-electricity cost at time step t. For the battery system, this cost is computed from the electricity price and the battery power exchanged with the grid, using the same price for purchases and sales; minimizing J_{grid} is equivalent to maximizing profit. For the cookie factory, it is computed from the grid power required by the heat-pump system after accounting for available wind power and is minimized directly. Thus, a lower value of J_{grid} indicates higher profit for the battery and lower operating cost for the cookie factory.

The cumulative reward is reported as a secondary metric:

$$R_{\mathrm{ep}} = \sum_{t=0}^{N-1} r_t.$$

This value is useful for analyzing the RL objective because it combines the economic term with penalty terms. However, it is not used as the only comparison criterion, since the reward may include artificial penalty weights that are not part

of the physical economic objective. Therefore, the economic objective and constraint satisfaction are evaluated separately.

Constraint satisfaction is measured by aggregating the violation terms recorded during the rollout. For a set of constraint residuals or violation magnitudes $v_{\ell,t}$, the total violation is defined as

$$V_{\text{tot}} = \sum_{t=0}^{N-1} \sum_{\ell} v_{\ell,t}.$$

In addition, the number or frequency of time steps with nonzero violations is considered:

$$f_{\text{viol}} = \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{I} \left(\sum_{\ell} v_{\ell,t} > \varepsilon \right),$$

where ε is a small numerical tolerance. For the battery system, the relevant checks include SOC bounds, power limits, and the terminal SOC requirement. For the cookie factory, the violation categories include bypass feasibility, complementarity of charging and discharging, TES temperature bounds, grid-power bounds, coupling residuals, and the terminal cyclic TES condition.

The evaluation also records trajectory-level operating indicators. For the BESS, these include the SOC trajectory, commanded and effective battery power, terminal SOC, and cumulative grid-side cost and the corresponding profit.

Finally, computation time is used to assess practical applicability. For mathematical optimization, the relevant value is the solver time required to compute a complete schedule for the evaluation horizon. For RL, the one-time training effort is reported separately from the online evaluation time. This distinction is important because RL can require substantial training effort, but a trained policy can usually produce setpoints much faster than solving a nonlinear optimization problem from scratch for every horizon.

5 Results and Discussion

The previous section described the RL formulation, training setup, and evaluation procedure for both use cases. This section presents and discusses the resulting policy performance relative to mathematical optimization. All computations were performed in Python 3.14.1, with Gymnasium 1.2.2 used to implement the reinforcement-learning environments and Stable-Baselines3 2.7.0, based on PyTorch 2.9.1, used to train the RL policies. Experiments were conducted on an AMD Ryzen 7 7840HS CPU with Radeon 780M Graphics.

5.1 Battery Energy Storage System

The battery case study was first used to compare different learning-rate schedules for SAC and TD3. Each configuration was trained and evaluated over 50 runs on the same battery environment and evaluation setup.

The linear schedule achieved the best mean reward and the lowest mean objective value among the tested configurations. However, the differences between the three schedules were small relative to the variation in objective values across runs. This indicates that the battery environment is not highly sensitive to the exact learning-rate decay shape in this experiment. The linear schedule was therefore used as a stable baseline for the subsequent comparison against mathematical optimization.

The trained SAC and TD3 policies were compared with mathematical optimization over 1000 randomly sampled 24-hour price windows. Table 2 summarizes the resulting objective values, computation times, and SOC ranges, using mathematical optimization as the baseline.

Metric	SAC policy	TD3 policy	Mathematical optimization
Mean objective [EUR]	-2.8911 ± 1.2408	-3.0223 ± 1.2208	-3.1851 ± 1.2212
Objective range [EUR]	$[-7.7304, -0.3483]$	$[-7.7841, -0.2604]$	$[-7.8530, -0.4353]$
Mean computation time [s]	0.0126 ± 0.0105	0.0082 ± 0.0067	0.5605 ± 0.6295
Time range [s]	$[0.0102, 0.3444]$	$[0.0070, 0.2150]$	$[0.1164, 9.2371]$
Mean maximum SOC	0.8661 ± 0.1620	0.8905 ± 0.1458	0.8793 ± 0.1595
Mean minimum SOC	0.2129 ± 0.1887	0.1947 ± 0.1776	0.1925 ± 0.1796

Table 2: Comparison between the trained SAC and TD3 policies and mathematical optimization over 1000 random 24-hour windows. Mathematical optimization serves as the baseline; lower (more negative) objective values correspond to higher profit.

Mathematical optimization achieved the lowest mean objective value and therefore the highest mean profit. It was the best-performing method overall. Among SAC and TD3, TD3 performed better. It achieved a lower mean objective value and higher mean profit, and was closer to the mathematical-optimization baseline. The mean objective gap was 0.1628 ± 0.1284 EUR for TD3 and 0.2215 ± 0.0811 EUR for SAC. Since the objective is minimized, both positive gaps indicate that the RL policies achieved lower profit than mathematical optimization. TD3 performed better in this relatively simple and deterministic environment. This could be explained by TD3 directly maximizing the discounted return, whereas SAC balances discounted-return maximization with maintaining policy entropy. Further TD3 results are presented below.

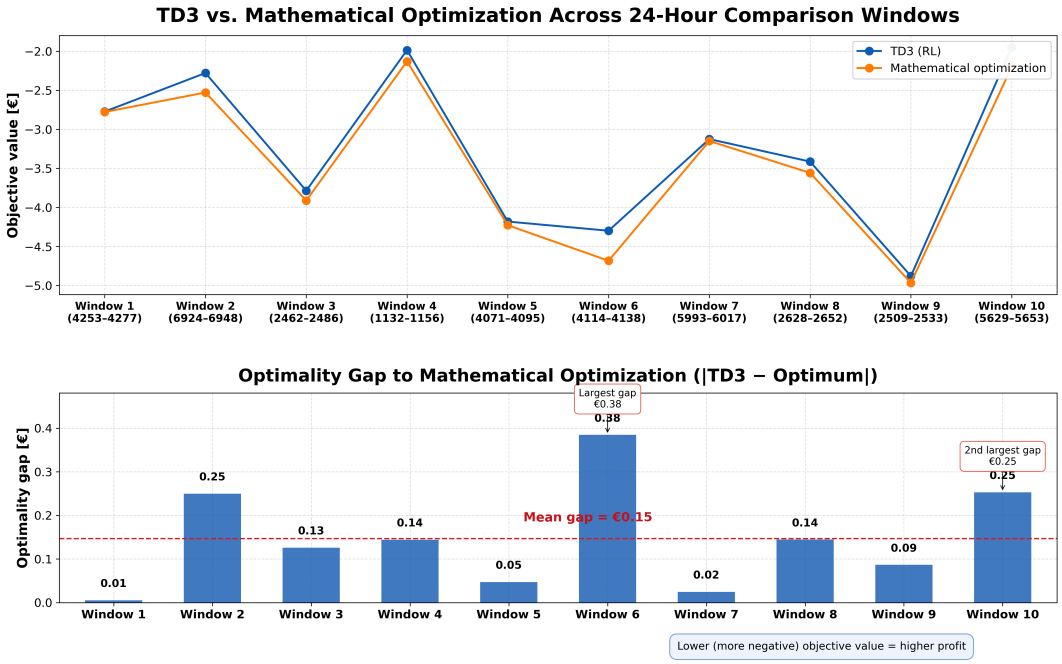


Figure 4: TD3 and mathematical-optimization objective values for ten randomly selected 24-hour comparison windows. The lower panel shows the corresponding TD3–mathematical-optimization gaps.

The small objective gaps shown in Figure 4 indicate that the TD3 policy closely

approaches the mathematical-optimization solution across the selected comparison windows.

The SOC trajectories are also similar in the current comparison. The TD3 policy reached a mean minimum SOC of 0.1947 and a mean maximum SOC of 0.8905, while the mathematical solution reached a mean minimum SOC of 0.1925 and a mean maximum SOC of 0.8793. This indicates that both methods used a comparable portion of the available storage flexibility.

Figure 5 shows an illustrative 24-hour dispatch trajectory for a TD3 policy and the mathematical optimization solution over the 24-hour horizon. The upper panel compares grid-power profiles and electricity price, showing that the TD3 actions closely follow the mathematical-optimization schedule. The lower panel compares SOC trajectories, both of which satisfy the terminal-SOC requirement.

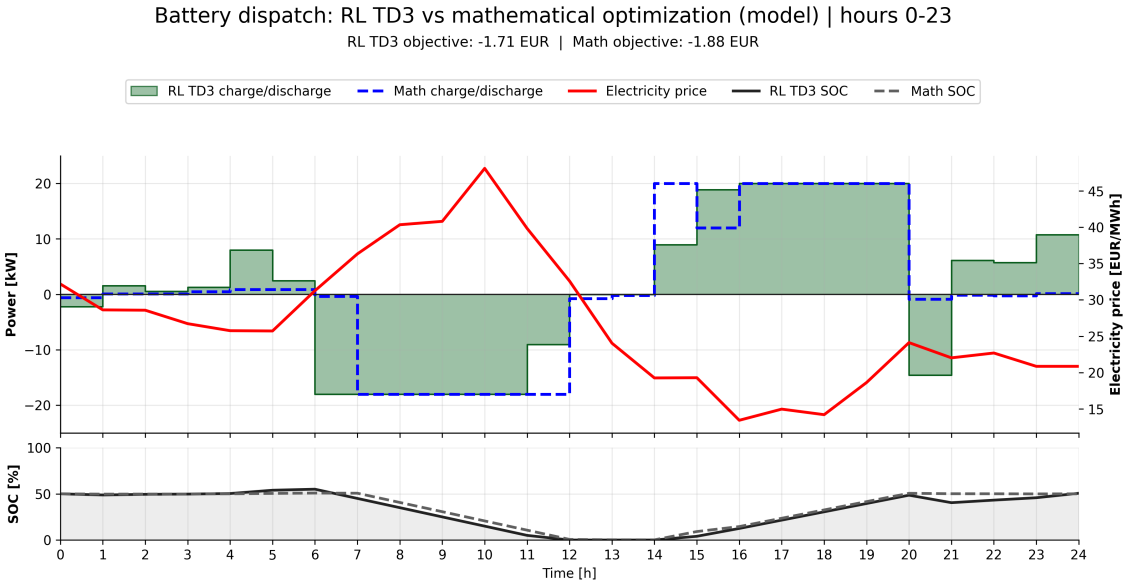


Figure 5: Illustrative 24-hour battery dispatch comparison between the TD3 policy and mathematical optimization.

In a separate feasibility evaluation, the TD3 policy without the progressive reachability penalty violated the terminal-SOC constraint in 4% of the evaluation windows. Adding the progressive penalty reduced the violation rate to zero but increased the mean objective gap between TD3 and mathematical optimization to 0.2436 ± 0.1473 EUR, illustrating the trade-off between strict terminal-constraint satisfaction and economic performance. Upon further investigation, the mean terminal-SOC shortfall without the progressive penalty was only 0.000252. Given its higher profit, this small shortfall was considered tolerable, and the TD3 policy without the progressive reachability penalty was selected as the final model.

The main advantage of the TD3 policy was computation time. The trained TD3 model required, on average, 0.0082 s to generate a 24-hour rollout, while mathematical optimization required 0.5605 s on average to solve the corresponding optimization problem. This corresponds to an average speedup of approximately $68.13\times$. For TD3, approximately one hour of offline training is required beforehand and is therefore not included in the online computation time. These results therefore illustrate the central trade-off observed in the battery case: RL provides very fast online setpoint generation after training and approaches the mathematical optimization result closely, while direct optimization still provides the best objective value.

5.2 Cookie Production Factory

5.3 Key Findings

6 Conclusion and Future Work

Summarizes the main findings and outlines possible future work.

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